

BOEKBESPREKINGS/BOOK REVIEWS

Grundbegriffe der Wahrscheinlichkeitstheorie

by K. Hinderer

First edition, VII + 247 pages, Springer-Verlag, Berlin, Heidelberg, New York, 1972.
DM 19,80; US \$6.30

This book has the immaculate quality of other German books on probability theory (e.g. by Richter, Krickeberg, Bauer). It contains less material than the three books mentioned and restricts itself to a short, but well-rounded foundation. Some of the more space consuming proofs are omitted, but precise references for these proofs are given, and numerous "remarks" and "complements" indicate directions for further study.

The book has 3 chapters of roughly equal length, the first of which deals with discrete probability spaces. The obvious advantage of this is that the student can develop a feeling for the problems and concepts before indulging in the theory of measure and integration (Chapter 2), which is necessary for the treatment of general probability spaces (Chapter 3). The second chapter contains sections on classes of sets, measures, extension of measures, measurable functions, the general integral and its properties, and measures in product spaces. The third deals with the general concept of the density, P-measures with Lebesgue densities, stochastic independence, expectations, conditional expectations, Laplace transforms, and characteristic functions, and gives an outlook on some stochastic processes.

In South Africa this would be a very suitable text for a course on probability theory in our Honours in Mathematical Statistics, possibly spread in short courses over the second, third and fourth year.

The style is very clear and there are almost no misprints. The print is small and the pages are cramped but these disadvantages could be rectified in an English edition for which, I am sure, we will not have to wait very long.

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STATISTICS: The Arithmetic of decision-making

by J.M. Leonard

First edition, 176 pages. The English Universities Press Ltd, London, 1971. U.K. £1.25

The author's purpose with this book is to meet the needs of students in elementary statistics at both school and university levels. He keeps away from mathematical statistics obviously to make the book more acceptable for those whose knowledge of mathematics

goes as far as a high school level. In many cases a little more explanation of the mathematical background would have made it easier to cope with some concepts. For example, it is noted that formal definitions of the sample median and mode in the case of grouped data is omitted.

On the whole it can be mentioned that the book is being read easily, and this, together with the omission of the mathematical tools being used, make it an acceptable textbook for students at an undergraduate level in the social sciences.

The author covers a wide range of topics. In the first chapter the general use of statistics is described as well as tabulation and graphical representation of sample values.

The second and third chapters deal with the average values and measures of dispersion of a sample.

In chapter 4 probability theory is treated in a very elementary way and in chapters 5 and 6 the binomial and Poisson distributions are introduced. The derivation of the mean and variance of these two distributions is treated unnecessarily because it may be too difficult for those students the author has in mind.

The use of the normal distribution (as an approximation for the binomial distribution, for example) is illustrated in chapter 7. A few more examples would have made this topic more clear.

In chapters 8 through 10 statistical inference is being dealt with and the common cases of testing of hypotheses such as testing for a proportion, mean, the difference between means and the difference between proportions are treated satisfactorily. Confidence intervals and the application thereof are discussed for the one sample case.

In chapters 11 and 12 the author deals with correlation and regression analysis and the inclusion of rank correlation may be specifically noted. The chi square distribution with its applications in the case of contingency tables is treated in chapter 13 and in the last chapter the brief discussion of miscellaneous topics such as probability functions, Sheppard's correction etc. concludes the book.

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Introduction to Probability and Statistics

by Henry L. Alder and Edward B. Roessler

Fifth edition, xii + 373 pages, W.H. Freeman and Company, San Francisco, 1972. Cloth £4.20, paper £2.80

The first edition of this book (which was reviewed by T.A. Bancroft in JASA, vol. 56, pp. 184-5) appeared in 1960. Since that first edition its volume increased by about fifty percent.

According to the authors the book was written to serve as a textbook for a one-semester introductory course in probability and statistics for students in natural and social sciences and certain humanities. The mathematical prerequisite is the equivalent of at least two years of high school algebra.

After a short historical introductory chapter, there follows chapters on summation

notation, organization and analysis of data, probability theory, the binomial and normal distributions, sampling theory, hypothesis testing for large samples, t-tests, nonparametric statistics, regression and correlation, chi-square tests, index numbers, time series, F-tests and analysis of variance. A good feature is the large number of well chosen exercises illustrating the use of statistics in most fields of specialization. Appropriate, easy to use tables and answers to odd-numbered exercises are given.

Improvements on the previous edition include the addition of a section on the Poisson distribution and sections on Duncan's Multiple Range Test and the randomized complete block design. The chapter on hypothesis testing has been improved considerably, though the treatment of confidence intervals remains unsatisfactory. Some 80 new exercises have been added.

In general, except for some awkward proofs and derivations, the book reads easily and is quite clear. The treatment of random variables and probability distributions is, however, inadequate and will probably not suffice all interested. Examples of awkward proofs/derivations are the proofs of theorems on probability in avoiding the use of elementary set theory and the derivation of the binomial variance using the general formula for a sample variance. The choice of topics included in the chapters on inference is acceptable.

In conclusion, this book is not an outstanding one, but as a textbook for an introductory service course in probability and statistics, the present edition can compete well with others available in this field.

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Statistische Auswertungsmethoden

by L. Sachs

Third revised edition, 545 pages, 58 figures and 189 tables. Springer-Verlag, Berlin, 1972. DM 58,—; U S \$ 18,10.

A review of the second edition of *Statistische Auswertungsmethoden* has appeared in 1970 in this journal — the review of the third edition can thus be concise.

In the new edition the presentation of the material has been improved in several ways by means of which many of the criticisms against the first two editions have been met. But in spite of this and notwithstanding its non-mathematical level, the book still could not be recommended for self-study or as a text book in an introductory course in statistical procedures for the non-mathematician. The computational aspects of practical applications of the procedures treated are over emphasised at the cost of an insufficient emphasis on conceptual aspects and the underlying principles of the different procedures. In many applications, for example, no or at least insufficient attention has been paid to the conditions under which the procedures are applicable.

The main feature of this book, however, is that it contains a surprisingly large number of statistical procedures, many of which will be new to the reader, and with them many

useful tables not usually found in statistical textbooks. Furthermore, the bibliography comprises 58 pages and contains many new and unknown references to the reviewer. To summarise, this book can be recommended as a useful reference book for the more experienced user of statistical techniques.

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